Interest Rate and Credit Risk Models
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Project 1

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1 Exercise 1

From the data, we first know that :

$$y(0,1) = 3\%, \quad y(0,2) = 3.8\%, \quad y(0,3) = 4.5\%.$$

1.a Zero-coupon bond prices

The zero-coupon bond price is given by

$$P(0,T) = e^{-y(0,T)T}.$$

Hence,

$$\begin{aligned} P(0,1) &= e^{-0.03\cdot 1} = 0.9704, \\ P(0,2) &= e^{-0.038\cdot 2} = 0.9268, \\ P(0,3) &= e^{-0.045\cdot 3} = 0.8737. \end{aligned}$$

1.b Simple forward rates

The simple forward rate for the period $[T_0, T_1]$ observed at time $t \leq T_0$ is

$$F(t, T_0, T_1) = \frac{1}{\delta}\left(\frac{P(t,T_0)}{P(t,T_1)} - 1\right),$$

where $\delta = T_1 - T_0$.

$$\begin{aligned} F(0,1,2) &= \frac{0.9704}{0.9268} - 1 = 0.0470, \\ F(0,2,3) &= \frac{0.9268}{0.8737} - 1 = 0.0608. \end{aligned}$$

1.c FRA valuation

The time-t value of the FRA can be expressed in terms of the forward rate as:

$$V_{\text{FRA}}(t, T_1, T_2) = N\, P(t, T_2)\, \delta\, [F(t, T_1, T_2) - K].$$

At time $t = 0$, this becomes:

$$V_{\text{FRA}}(0, T_1, T_2) = N\, P(0, T_2)\, (T_2 - T_1)\, [F(0, T_1, T_2) - K].$$

Substituting the given values:

$$\begin{aligned} V_{\text{FRA}}(0,1,2) &= 100\,\text{m} \times 0.9268 \times (2-1) \times (0.0470 - 0.0400) \\ &= 100\,\text{m} \times 0.9268 \times 0.0070 \\ &= 0.648\,\text{million } \$. \end{aligned}$$

The FRA is more valuable to the long (borrower), because the forward rate $F(0,1,2) = 0.047$ exceeds the contract rate $K = 0.04$, resulting in a positive value of $V_{FRA} = 0.648$ million.

1.d Parallel upward shift of 50 bps

After a parallel upward shift of 50 basis points (0.50%), the new zero-coupon yields become:

$$y(0,1) = 3.5\%, \qquad y(0,2) = 4.3\%, \qquad y(0,3) = 5.0\%.$$

The corresponding zero-coupon bond prices are:

$$P'(0,1) = e^{-0.035} = 0.9656, \qquad P'(0,2) = e^{-0.043\times 2} = 0.9176.$$

The new forward rate is therefore:

$$F'(0,1,2) = \frac{P(0,1)}{P(0,2)} - 1 = 0.0523 = 5.23\%.$$

Hence, the new FRA value is:

$$V'_{\text{FRA}}(0,T_1,T_2) = N\,P(0,T_2)\,(T_2 - T_1)\,[F(0,1,2) - K].$$

Substituting the given values:

$$V'_{\text{FRA}}(0,1,2) = 100\,\text{m} \times 0.9176 \times (2-1) \times (0.0523 - 0.04) = 1.129\,\text{million } \$.$$

—

The duration is defined as:

$$\text{Duration} = -\frac{1}{V_{\text{FRA}}(0)}\,\frac{dV_{\text{FRA}}(s)}{ds}\bigg|_{s=0},$$

where the zero-coupon bond price is expressed as:

$$P(0,T) = e^{-y(0,T)\,T}.$$

A parallel shift means that each yield increases by the same small amount s:

$$y_s(0,T) = y(0,T) + s.$$

Thus, the shifted discount factor becomes:

$$P_s(0,T) = e^{-y_s(0,T)T} = e^{-(y(0,T)+s)T} = e^{-y(0,T)T}\,e^{-sT} = P(0,T)\,e^{-sT}.$$

—

The time-0 value of a payer FRA is:

$$V_{\text{FRA}}(0) = N\left[P(0,T_1) - (1+K\Delta)P(0,T_2)\right]. \tag{1}$$

Replacing $P(0,T_1)$ and $P(0,T_2)$ by their shifted versions gives:

$$V_{\text{FRA}}(s) = N\left[P(0,T_1)e^{-sT_1} - (1+K\Delta)P(0,T_2)e^{-sT_2}\right]. \tag{2}$$

Differentiating $V_{\text{FRA}}(s)$ with respect to s, we obtain:

$$\frac{dV_{\text{FRA}}}{ds} = N\left[-T_1P(0,T_1)e^{-sT_1} + (1+K\Delta)T_2P(0,T_2)e^{-sT_2}\right]. \tag{3}$$

For $s = 0$:

$$\frac{dV_{\text{FRA}}}{ds}\bigg|_{s=0} = N\left[-T_1P(0,T_1) + (1+K\Delta)T_2P(0,T_2)\right]. \tag{4}$$

Substituting the numerical values:

$$\begin{aligned}
\frac{dV_{\text{FRA}}}{ds}\bigg|_{s=0} &= 100{,}000{,}000\left[-1 \times 0.9704 + (1 + 0.04 \times 1) \times 2 \times 0.9268\right] && (5)\\
&= 100{,}000{,}000\left[-0.9704 + 1.927744\right] && (6)\\
&= 100{,}000{,}000 \times 0.957344 && (7)\\
&= 95{,}734{,}400. && (8)
\end{aligned}$$

$$\boxed{\left.\frac{dV_{\text{FRA}}}{ds}\right|_{s=0} = 95.73 \text{ million.}}$$

Finally, the duration is:

$$\text{Duration} = -\frac{1}{V_{\text{FRA}}(0)} \left.\frac{dV_{\text{FRA}}(s)}{ds}\right|_{s=0} = -\frac{95.73\,\text{million}}{0.648\,\text{million}} = -148.$$

—

The computed duration is -148, which is negative. A negative duration indicates that the FRA value moves in the **same** direction as interest rates. This is consistent with the behavior of a payer FRA, where the holder receives the floating rate and pays the fixed rate: an upward parallel shift in the yield curve (i.e., higher expected future rates) increases the expected payoff of the FRA, thereby increasing its present value.

In other words, the negative sign of the duration shows that the FRA's value is positively correlated with yields — when interest rates rise by 50 basis points, the FRA value increases proportionally to its duration magnitude.

1.e Second derivative (convexity) under a parallel shift

Differentiating the FRA value a second time with respect to the parallel shift s, we obtain:

$$\frac{d^2V_{\text{FRA}}(s)}{ds^2} = N\left[T_1^2 P(0,T_1)e^{-sT_1} - (1+K\Delta)T_2^2 P(0,T_2)e^{-sT_2}\right]. \tag{9}$$

Evaluating at $s = 0$ gives:

$$\left.\frac{d^2V_{\text{FRA}}}{ds^2}\right|_{s=0} = N\left[T_1^2 P(0,T_1) - (1+K\Delta)T_2^2 P(0,T_2)\right]. \tag{10}$$

Substituting the given numerical values:

$$\begin{aligned}
\left.\frac{d^2V_{\text{FRA}}}{ds^2}\right|_{s=0} &= 100{,}000{,}000\left[1^2 \times 0.9704 - (1+0.04)\times 2^2 \times 0.9268\right] && (11)\\
&= 100{,}000{,}000\left[0.9704 - 1.04 \times 4 \times 0.9268\right] && (12)\\
&= 100{,}000{,}000\left[0.9704 - 3.855488\right] && (13)\\
&= 100{,}000{,}000 \times (-2.885088) && (14)\\
&= -288{,}508{,}800. && (15)
\end{aligned}$$

$$\boxed{\left.\frac{d^2V_{\text{FRA}}}{ds^2}\right|_{s=0} = -2.88 \times 10^8}$$

—

The convexity of the FRA is defined as:

$$\text{Convexity} = \frac{1}{V_{\text{FRA}}(0)} \left.\frac{d^2V_{\text{FRA}}(s)}{ds^2}\right|_{s=0}.$$

Substituting the numerical value, we find:

$$\begin{aligned}
\text{Convexity} &= \frac{-2.88 \times 10^8}{0.648 \times 10^6}\\
&= -444
\end{aligned}$$

$$\boxed{\text{Convexity} = -444}$$

—

The analytical result shows that the second derivative of the FRA value with respect to a parallel shift in the yield curve is **negative**, indicating that FRAs exhibit **negative convexity**. This means that the sensitivity of the FRA value to yield changes diminishes as rate movements grow larger — in other words, the relationship between FRA value and yield shifts is nonlinear, and the value change becomes progressively smaller for larger yield variations.

1.f Non-parallel shift (steepening)

We keep $y(0,2)$ fixed and apply a non-parallel move: $y(0,1) \downarrow 20$ bps and $y(0,3) \uparrow 20$ bps, i.e.

$$y(0,1) = 2.8\%, \qquad y(0,2) = 3.8\%, \qquad y(0,3) = 4.7\%.$$

and $P(0,T) = e^{-y(T)\,T}$, so

$$P(0,1) = e^{-0.028} = 0.9724, \qquad P(0,2) = e^{-0.038\cdot 2} = 0.9268.$$

Using the forward-rate definition,

$$F(0,1,2) = \frac{P(0,1)}{P(0,2)} - 1 = \frac{0.9724}{0.9268} - 1 = 0.0492 = 4.92\%.$$

we can the compute

$$V_{\text{FRA}^{new}}(0) = N\left[P(0,T_1) - (1+K\Delta)P(0,T_2)\right]. = 100m\left[0.9724 - (1+0.04*1)0.9268\right]. = 0,85 million \tag{16}$$

Only $y(0,1)$ and $y(0,2)$ enter $F(0,1,2)$, so the change in $y(3)$ is irrelevant here. Compared to the base case $F(0,1,2) = 4.7\%$, the steepening raises F to 4.92%. Since in (c) the FRA has $K = 4\%$ and the long (borrower) benefits when $F > K$, the FRA holder from (c) benefits from this steepening (the contract's value increases).

2 Exercise 2

a)
Knowing the following quarterly forward rates, expressed as simple annually: F(0,0,0.25), F(0,0.25,0.5), F(0,0.5,0.75), F(0,0.75,1), we can compute the corresponding zero-coupon bond prices using:

$$\frac{1}{P(t,T)} = 1 + (T-t)F(0,t,T) \Rightarrow P(t_i,t_{i+1}) = \frac{1}{1+\delta F(0,t_i,t_{i+1})}$$

We also know that

$$P(t_i,t_{i+1}) = \frac{P(0,t_{i+1})}{P(0,t_i)} \quad \Rightarrow P(0,t_{i+1}) = P(0,t_i) * P(t_i,t_{i+1}) = \frac{P(0,t_i)}{1+\delta F(0,t_i,t_{i+1})}$$

The results are as follows.

$$P(0,0.25) = \frac{1}{1+\delta F(0,0,0.25)} = \frac{1}{1+0.25*0.048} = 0.9881$$
$$P(0,0.5) = \frac{P(0,0.25)}{1+\delta F(0,0.25,0.5)} = \frac{0.9881}{1+0.25*0.05} = 0.9759$$
$$P(0,0.75) = \frac{P(0,0.5)}{1+\delta F(0,0.5,0.75)} = \frac{0.9759}{1+0.25*0.053} = 0.9632$$
$$P(0,1) = \frac{P(0,0.75)}{1+\delta F(0,0.75,1)} = \frac{0.9632}{1+0.25*0.056} = 0.9499$$

b)

$$R_{swap}(0) = \frac{1 - P(0,1)}{\delta \sum_{i=1}^{4} P(0, i/4)} = \frac{1 - 0.9499}{0.25 * (0.9881 + 0.9759 + 0.9632 + 0.9499)} = 0.0517$$

c)

We want to show that:

$$R_{swap}(0) = \sum_{i=1}^{4} w_i F(0, (i-1)/4, i/4), \quad w_i = \frac{\delta P(0, i/4)}{\delta \sum_{j=1}^{4} P(0, j/4)}.$$

Starting with:

$$R_{swap}(0) = \sum_{i=1}^{4} w_i F(0, (i-1)/4, i/4) = \sum_{i=1}^{4} \frac{\delta P(0, i/4)}{\delta \sum_{j=1}^{4} P(0, j/4)} F(0, (i-1)/4, i/4)$$

$$\text{We know that:} \quad \delta F(0, t_i, t_{i+1)} = \frac{P(0, t_i)}{P(0, t_{i+1})} - 1 = \frac{P(0, (i-1)/4)}{P(0, i/4)} - 1,$$

So we can replace and write:

$$\begin{aligned}
R_{swap}(0) &= \frac{1}{\delta \sum_{j=1}^{4} P(0, j/4)} \sum_{i=1}^{4} P(0, i/4) \ (\frac{P(0, (i-1)/4)}{P(0, i/4)} - 1) \\
&= \frac{1}{\delta \sum_{j=1}^{4} P(0, j/4)} \sum_{i=1}^{4} P(0, (i-1)/4) - P(0, i/4) \\
&= \frac{1}{\delta \sum_{j=1}^{4} P(0, j/4)} (P(0,0) - P(0,1)) \\
&= \frac{1 - P(0,1)}{\delta \sum_{j=1}^{4} P(0, j/4)}
\end{aligned}$$

We now compute w_i, using:

$$P(0, 0.25) = 0.9881, \quad P(0, 0.5) = 0.9759, \quad P(0, 0.75) = 0.9632, \quad , \quad P(0,1) = 0.9499, \quad \sum_{j=1}^{4} P(0, j/4) = 3.8771$$

we have:

$$w_1 = \frac{0.9881}{3.8771} = 0.2549, \quad w_2 = \frac{0.9759}{3.8771} = 0.2517, \quad w_3 = \frac{0.9632}{3.8771} = 0.2484, \quad w_4 = \frac{0.9499}{3.8771} = 0.2450$$

Each w_i represents the relative present value weight of the i-th cash flow, meaning that earlier payments (with higher $P(0, t_i)$) have a slightly larger influence on the swap rate.

d)

We use the identity $\max(x, a) = a + (x - a)^+$. The PV of the floored floating leg is

$$\begin{aligned}
\mathrm{PV}_0(\text{floored float}) &= \sum_{i=1}^{4} P(0, t_i)\, \mathbb{E}_0^{\mathbb{Q}}\big[\delta\, \max(L_{i-1}, \kappa)\big] \\
&= \sum_{i=1}^{4} P(0, t_i) \left(\delta\, \kappa + \mathbb{E}_0^{\mathbb{Q}}\big[\delta\, (L_{i-1} - \kappa)^+\big]\right) \\
&= \delta\, \kappa \sum_{i=1}^{4} P(0, t_i) \;+\; \mathrm{PV}_0\big(\text{cap with strike } \kappa\big).
\end{aligned}$$

Let R_κ be the par fixed rate of the swap with this floored floating leg. The fixed leg pays $\delta\, R_\kappa$ at each t_i, so

$$\mathrm{PV}_0(\text{fixed}) = \delta\, R_\kappa \sum_{i=1}^{4} P(0, t_i).$$

Setting PVs equal gives

$$\delta\, R_\kappa \sum_{i=1}^{4} P(0, t_i) = \delta\, \kappa \sum_{i=1}^{4} P(0, t_i) + \mathrm{PV}_0(\text{cap with strike } \kappa) \quad \Rightarrow \quad R_\kappa \;=\; \kappa + \frac{\mathrm{PV}_0(\text{cap with strike } \kappa)}{\delta \sum_{i=1}^{4} P(0, t_i)}.$$

$R_\kappa \geq R_{swap}(0)$ as you must pay a higher fixed rate to compensate for receiving a protected floating leg (which cannot go below κ).

$$R_\kappa \;\geq\; R_{\text{swap}}(0).$$

3 Exercise 3

a) NS and NSS Curve Fitting (July 2025) :

We fit both the Nelson–Siegel (NS) and Nelson–Siegel–Svensson (NSS) models to the observed zero-coupon yields for July 2025 using non-linear least squares estimation. The parameter estimates and corresponding root mean squared errors (RMSE) are reported below.

Table 1: Estimated Parameters and Fit Quality (July 2025)

Model	β_0	β_1	β_2	β_3	λ_1	λ_2	RMSE
Nelson–Siegel (NS)	2.4375	1.1361	1.3269	–	7.2272	–	0.00555
Nelson–Siegel–Svensson (NSS)	2.5377	0.2843	1.9652	2.4063	0.8501	5.0802	0.00361

Figure 1 below displays the observed yields along with the fitted NS and NSS curves. Both models capture the overall shape of the yield curve, but the NSS specification provides a slightly better fit, as indicated by its lower RMSE.

- The NS model explains most of the yield curve shape using three parameters $(\beta_0, \beta_1, \beta_2)$ and one decay factor λ_1.
- The NSS model introduces an additional curvature term (β_3, λ_2), allowing more flexibility in matching the medium-term segment of the curve.

Interpretation. The NSS model achieves a lower RMSE (0.0036 vs. 0.0055), implying a better in-sample fit. This improvement comes from the second exponential term, which enhances the model's ability to capture subtle curvature around intermediate maturities. However, the gain in accuracy is relatively modest compared to the added complexity of the NSS specification.

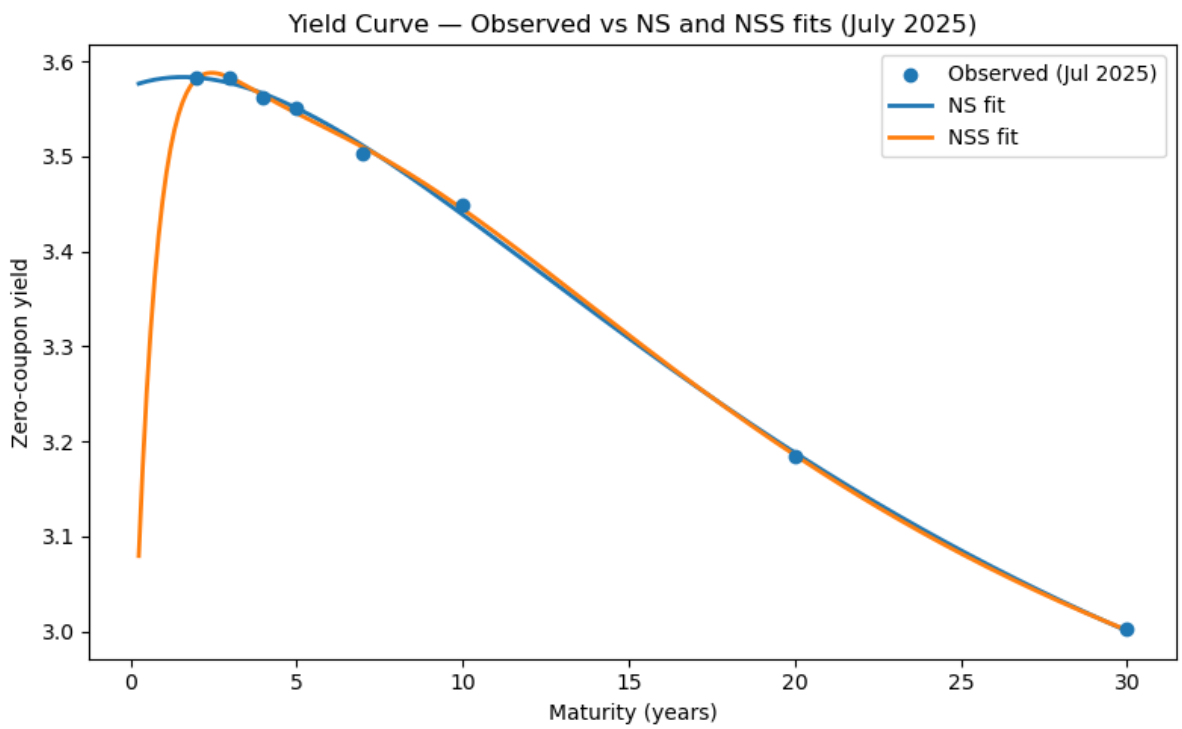

Figure 1: Observed vs. Fitted Yield Curves (NS and NSS, July 2025)

b) PCA of Monthly Yield Changes (Aug 2015 – Jul 2025) :

We compute monthly yield changes

$$\Delta y_t(T) = y_t(T) - y_{t-1}(T)$$

for maturities $T = 2, 3, 4, 5, 7, 10, 20, 30$ years, using monthly zero-coupon yields between August 2015 and July 2025.

Empirical covariance matrix. Table 2 shows the covariance matrix of monthly yield changes. Correlations are high and positive across maturities, indicating strong comovement.

Table 2: Covariance matrix of monthly yield changes (%)

	2y	3y	4y	5y	7y	10y	20y	30y
2y	0.1991	0.1980	0.1940	0.1933	0.1894	0.1840	0.0923	0.0949
3y	0.1980	0.1974	0.1932	0.1924	0.1885	0.1832	0.0915	0.0942
4y	0.1940	0.1932	0.1897	0.1885	0.1848	0.1796	0.0899	0.0924
5y	0.1933	0.1924	0.1885	0.1880	0.1840	0.1788	0.0889	0.0916
7y	0.1894	0.1885	0.1848	0.1840	0.1807	0.1752	0.0878	0.0903
10y	0.1840	0.1832	0.1796	0.1788	0.1752	0.1706	0.0851	0.0876
20y	0.0923	0.0915	0.0899	0.0889	0.0878	0.0851	0.0669	0.0605
30y	0.0949	0.0942	0.0924	0.0916	0.0903	0.0876	0.0605	0.0570

Principal components. The first three eigenvalues and their explained variances are shown below.

- **PC1 – Level factor:** all loadings have the same sign, representing parallel shifts of the yield curve.
- **PC2 – Slope/curvature factor:** short-term and long-term loadings have opposite signs, capturing steepening or flattening.

Table 3: PCA eigenvalues and explained variance

Component	Eigenvalue	Variance (%)	Cumulative (%)
PC1	1.2245	97.19	97.19
PC2	0.0337	2.68	99.86
PC3	0.0004	0.03	99.89

Table 4: PCA loadings by maturity

Maturity	PC1	PC2	PC3
2y	0.404	−0.103	0.037
3y	0.403	−0.116	−0.409
4y	0.395	−0.107	0.254
5y	0.393	−0.129	−0.086
7y	0.385	−0.101	0.678
10y	0.374	−0.107	−0.421
20y	0.194	0.798	0.193
30y	0.197	0.538	−0.293

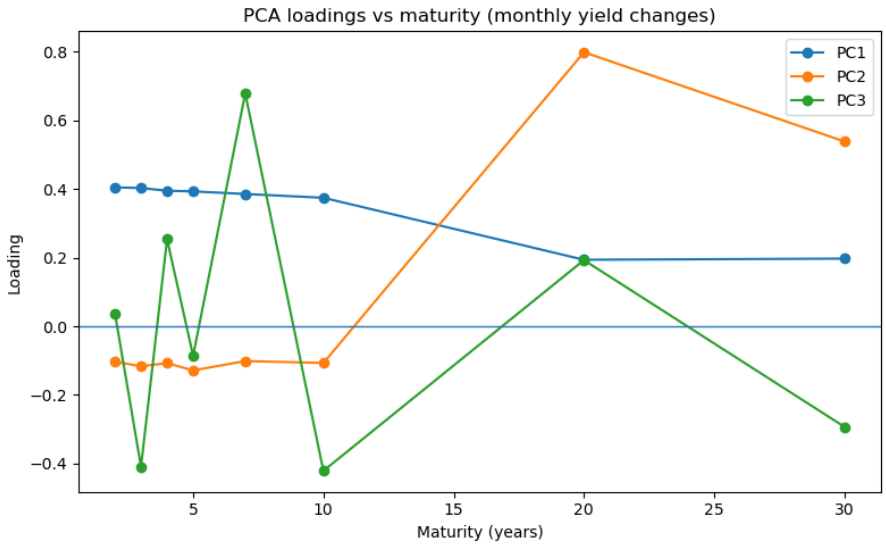

Figure 2: PCA loadings vs. maturity (monthly yield changes, Aug 2015–Jul 2025)

- **PC3 – Curvature (belly) factor:** changes sign around medium maturities, reflecting local curvature adjustments.

Together, the first two components explain about **99.9%** of total variance, confirming that yield curve dynamics are dominated by level and slope movements.

c) Comparison between NS Factors and Empirical PCA Loadings :

The comparison shows that the NS factors capture the broad structure of yield curve movements, but only the first factor aligns closely with empirical reality.

The first PCA loading (PC1) and the NS level factor match well : both are roughly flat across maturities and represent overall parallel shifts of the yield curve.

However, the second (PC2) and third (PC3) differ noticeably from the NS slope and curvature factors. The NS slope declines smoothly with maturity, but the empirical PC2 is much more irregular and even changes sign, indicating that real-world "slope" movements combine multiple effects, not just short–long rate differentials. Similarly, the NS curvature shows a clean hump shape, while PC3 fluctuates erratically, suggesting more complex and noisy medium-term dynamics in the data.

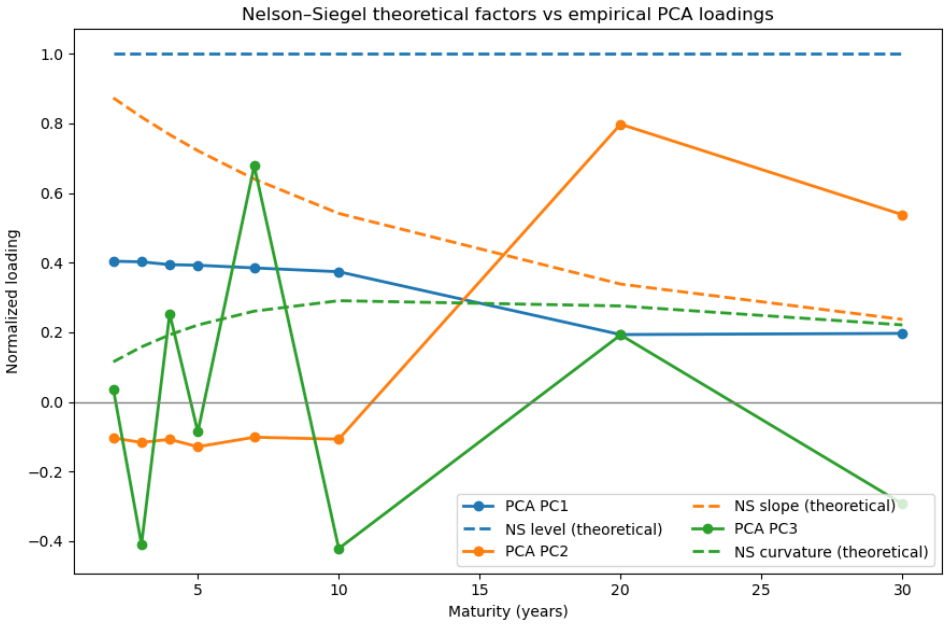

Figure 3: Nelson–Siegel theoretical factors vs. empirical PCA loadings

Nevertheless, the economic interpretation of the NS components remains valid:

- $I_0(T)$ corresponds to the **level** factor, describing uniform shifts in yields across maturities;
- $I_1(T)$ corresponds to the **slope** factor, capturing the difference between short- and long-term yields;
- $I_2(T)$ represents the **curvature** factor, reflecting changes concentrated around intermediate maturities.

4 Exercise 4

4.a Risk neutral dynamics

Under the risk neutral measure $\mathbb{Q}$, we have :

$$dW_t^* = dW_t + adt$$

Where a is the market price of risk and W_t^* is a $\mathbb{Q}$-brownian motion.
Therefore by replacing the $\mathbb{P}$-Brownian motion inside the short rate dynamics, we have :

$$dr_t = r_t(ac - b + (1-a)r_t)dt + r_t(c - r_t)(dW_t^* - adt)$$

$$dr_t = r_t(r_t - b)dt + r_t(c - r_t)dW_t^*$$

4.b Explicit expression for $D(\tau)$

Since there is absence of arbitrage we have that the discounted T-bond price is a martingale under the risk neutral measure $\mathbb{Q}$.
Let's start by computing the dynamics of the discounted T-bond price :

$$d(\frac{P(t,T)}{B(t)}) = \frac{1}{B(t)}dP(t,T) - \frac{1}{B(t)^2}P(t,T)dB(t)$$

Where we have :

$$\begin{aligned}
dP(t,T) &= r_t dD(\tau) + D(\tau)dr_t \\
&= r_t dD(\tau) + r_t D(\tau)((r_t - b)dt + (c - r_t)dW_t^*) \\
&= -r_t D'(\tau)dt + r_t D(\tau)(r_t - b)dt + r_t(c - r_t)D(\tau)dW_t^* \\
&= r_t(D(\tau)(r_t - b) - D'(\tau))dt + r_t(c - r_t)D(\tau)dW_t^*
\end{aligned}$$

Where we used :

$$dD(\tau) = -D'(\tau)dt$$

Therefore, by using :

$$dB(t) = B_t r_t dt$$

We have

$$\begin{aligned}
d(\frac{P(t,T)}{B(t)}) &= \frac{1}{B(t)}(r_t(D(\tau)(r_t - b) - D'(\tau))dt + r_t(c - r_t)D(\tau)dW_t^*) - \frac{1}{B(t)}r_t(1 + D(\tau)r_t)dt \\
&= \frac{r_t}{B(t)}(-(D(\tau)b + D'(\tau) + 1)dt + (c - r_t)D(\tau)dW_t^*)
\end{aligned}$$

For $\frac{P(t,T)}{B(t)}$ to be a $\mathbb{Q}$-martingale, we need the drift to be equal to 0 under the risk neutral measure. Therefore, the ODE can be expressed as :

$$D'(\tau) + D(\tau)b + 1 = 0$$

Solving the ODE

Multiply both sides by $e^{b\tau}$ and then integrate from 0 to τ

$$\begin{aligned}
e^{b\tau}D(\tau)b + D'(\tau)e^{b\tau} &= -e^{b\tau} \\
\frac{d}{d\tau}(e^{b\tau}D(\tau)) &= -e^{b\tau} \\
d(e^{b\tau}D(\tau)) &= -e^{b\tau}d\tau \\
e^{b\tau}D(\tau) - 0 &= \frac{1}{b}(1 - e^{b\tau}) \\
D(\tau) &= \frac{1}{b}(e^{-b\tau} - 1)
\end{aligned}$$

This reuslt is valid for $b > 0$.

Limit as $b \to 0$. To include the case $b = 0$, take the limit of the $b > 0$ solution:

$$D(\tau) = \frac{e^{-b\tau} - 1}{b}.$$

Let $x = -b\tau$. Then

$$\frac{e^{-b\tau} - 1}{b} = \frac{e^x - 1}{x}(-\tau) \xrightarrow[b\to 0]{} 1 \cdot (-\tau) = -\tau.$$

Therefore, in the limit $b \to 0$ we obtain

$$\boxed{D(\tau) = -\tau}.$$

Equivalently, when $b = 0$ the ODE becomes $D'(\tau) + 1 = 0$ with $D(0) = 0$, whose solution is also $D(\tau) = -\tau$.

4.c Economic interpretation

Why an affine bond price works. Under the risk–neutral measure $\mathbb{Q}$, a zero-coupon price $F(t,r) = P(t,T)$ solves

$$\partial_t F + \mu(r)\,\partial_r F + \tfrac{1}{2}\sigma^2(r)\,\partial_{rr} F - r\,F \;=\; 0, \qquad F(T,r) = 1,$$

with $\mu(r) = r(r-b)$ and $\sigma(r) = r(c-r)$. Using the affine (here linear) ansatz

$$P(t,T) = 1 + D(\tau)\,r, \qquad \tau = T - t,$$

we have $F_{rr} = 0$, so the diffusion term vanishes and the PDE reduces to

$$\big(D'(\tau) + b\,D(\tau) + 1\big)\,r = 0 \quad \Longrightarrow \quad D'(\tau) + b\,D(\tau) + 1 = 0,\; D(0) = 0,$$

whose solution is

$$D(\tau) = \frac{e^{-b\tau} - 1}{b} \quad (b > 0), \qquad \text{and} \qquad D(\tau) = -\tau \;\text{ when } b = 0.$$

Thus an affine price is consistent because the pricing operator maps linear functions of r back to linear functions (no curvature term).

Positivity and term-structure behavior. Assume $r_t \in [0,c]$ (nonnegative short rate, with volatility $r(c-r)$ vanishing at the boundaries 0 and c). Since $D(\tau) \le 0$ for all $\tau \ge 0$,

$$P(t,T) = 1 + D(\tau)\,r_t \in \big[\,1 + c\,D(\tau),\; 1\,\big].$$

For $b > 0$, $D(\tau) \in [-1/b, 0]$, hence

$$P(t,T) \in \big[\,1 - \tfrac{c}{b},\; 1\,\big].$$

Therefore $P(t,T) \ge 0$ for all maturities provided

$$\boxed{b > 0 \quad \text{and} \quad c \le b}.$$

Moreover, $D'(\tau) = -1 - bD(\tau) < 0$ for $\tau > 0$ when $b > 0$, so $P(t,T)$ is decreasing in T, as expected.

If $b = 0$, then $D(\tau) = -\tau$ and $P(t,T) = 1 - \tau r_t$ becomes negative for large maturities unless $r_t = 0$, which is economically undesirable. This motivates the conditions $b > 0$ and $c \le b$ to ensure positive and well-behaved bond prices.